Name: Nelson Puthuraja Nadar **Class:** S.E / INFT-B **Batch:** B-1 **Roll no.:** 21

Q.1] What is Jenkins?

Jenkins is an open source automation server written in Java that is widely used to implement Continuous Integration and Continuous Delivery in modern software development environments. It automates key stages of the Software Development Life Cycle by continuously monitoring version control systems such as Git, triggering automatic builds whenever code changes are committed, executing test cases, generating reports, and supporting deployment processes. Jenkins operates through a server based architecture and supports a distributed master and agent model, where the master manages scheduling and configuration while agents execute build jobs, thereby improving scalability and performance. It provides pipeline as code functionality through a Jenkinsfile, enabling developers to define build, test, and deployment workflows programmatically. With extensive plugin support, platform independence across Windows, Linux, and macOS, and integration with tools such as Maven, Docker, and Kubernetes, Jenkins enhances collaboration, ensures early detection of defects, reduces manual effort, and contributes to faster, reliable, and high quality software delivery in DevOps practices.

Q.2] Why Use Jenkins?

1. **Automation of Build Process**
   Jenkins automates the compilation and packaging of source code, reducing manual effort and minimizing human errors.
2. **Support for Continuous Integration**
   It enables frequent integration of code changes into a shared repository, ensuring that errors are detected early in the development cycle.
3. **Early Bug Detection**
   Automated testing after every build helps in identifying defects at an early stage, thereby improving software quality.
4. **Time Efficiency**
   By automating repetitive tasks such as building, testing, and deployment, Jenkins speeds up the overall development process.
5. **Improved Collaboration**
   It provides instant feedback to developers regarding build status, helping teams coordinate effectively.
6. **Extensive Plugin Support**
   Jenkins supports a large number of plugins, allowing integration with tools like Git, Maven, Docker, and other DevOps technologies.
7. **Platform Independence**
   Being Java based, Jenkins can run on multiple operating systems such as Windows, Linux, and macOS.
8. **Scalability Through Distributed Architecture**
   Its master and agent architecture allows distributed builds, improving performance and handling large scale projects efficiently.
9. **Continuous Delivery and Deployment Support**
   Jenkins facilitates automated deployment processes, ensuring faster and reliable software releases.

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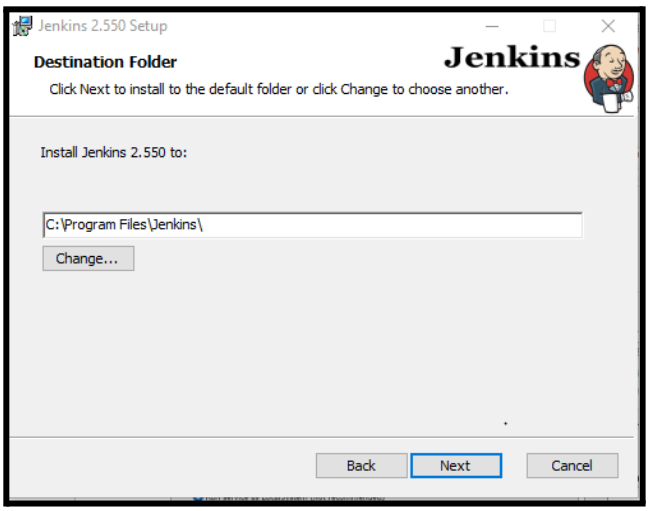
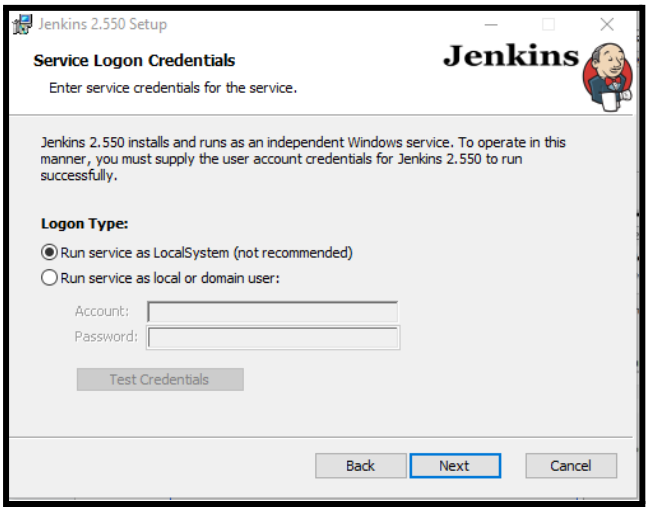
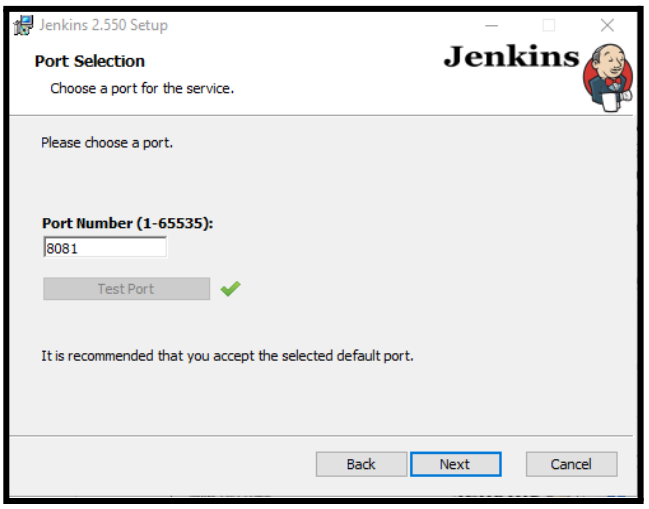
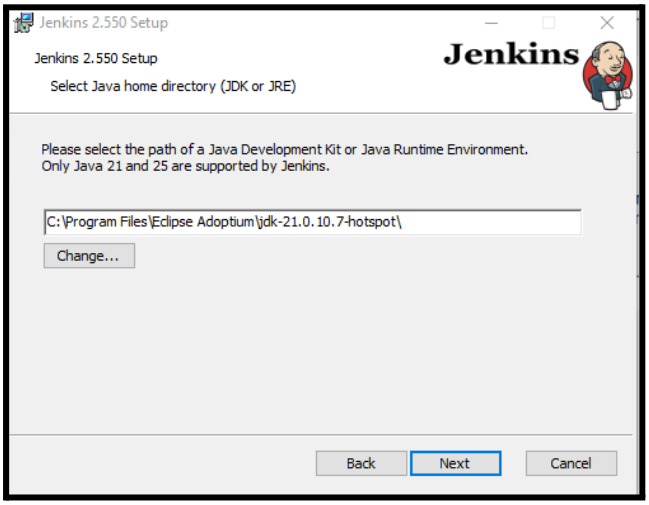
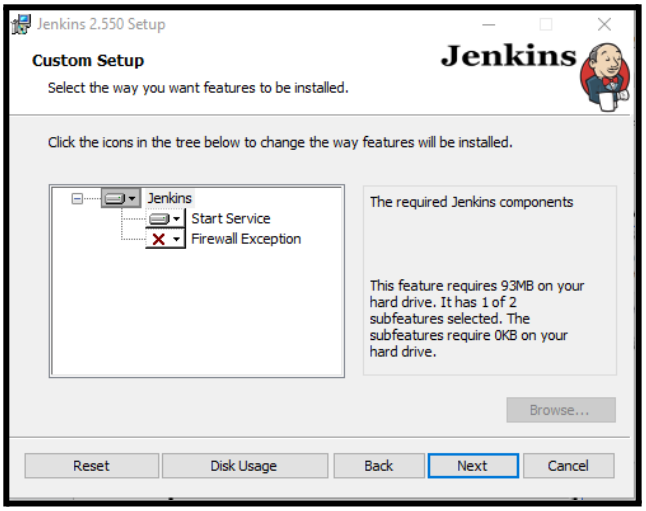
Install Jenkins











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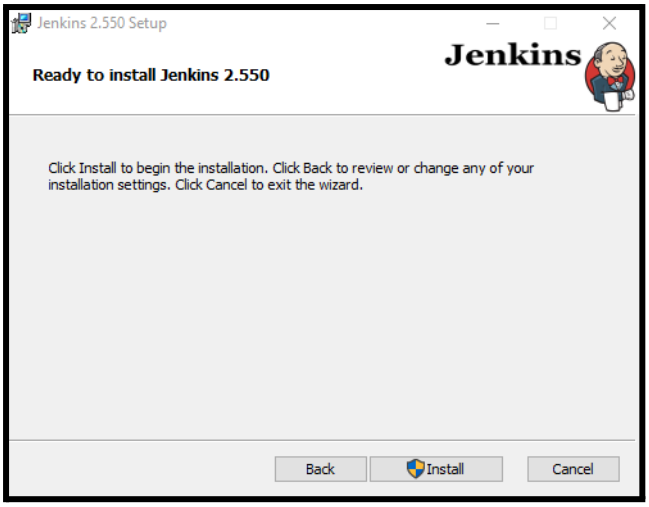

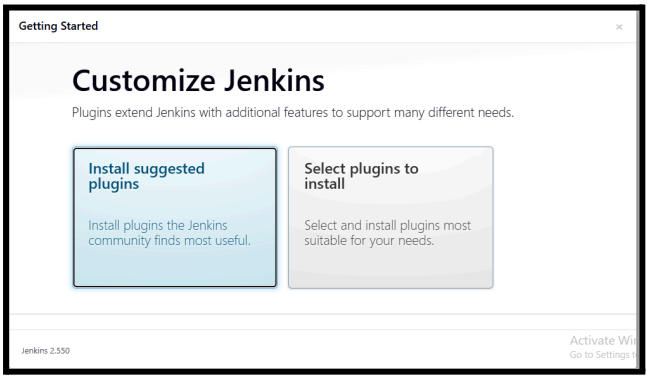

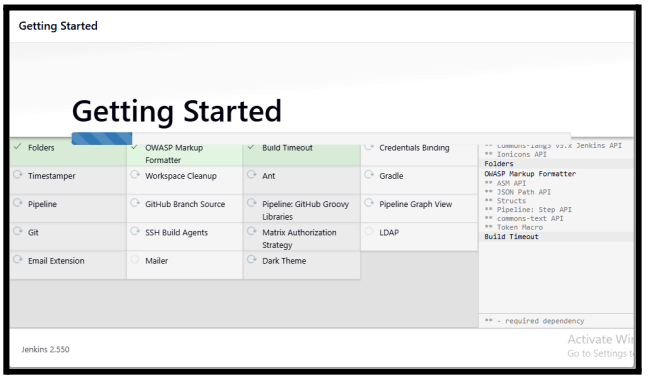

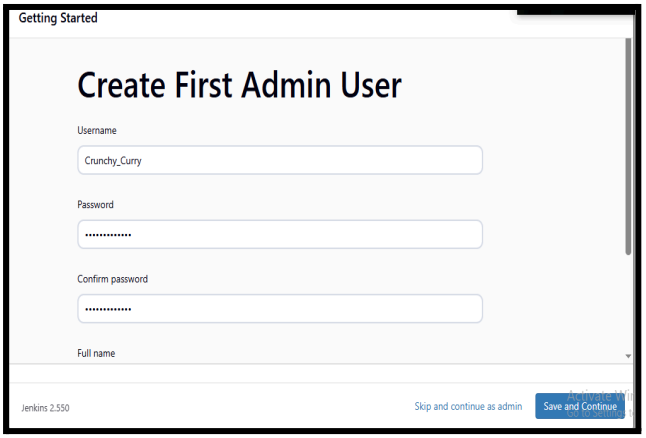

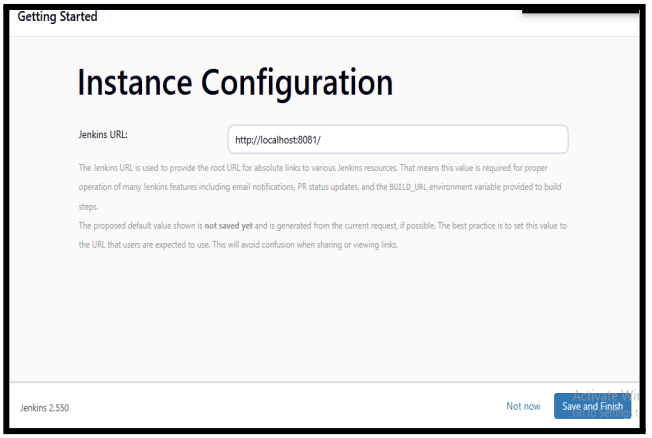

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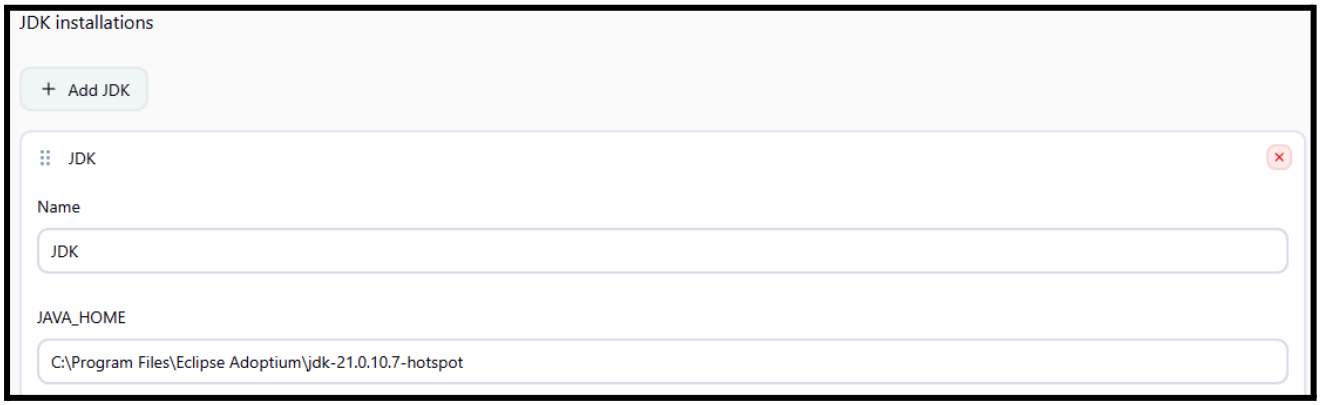
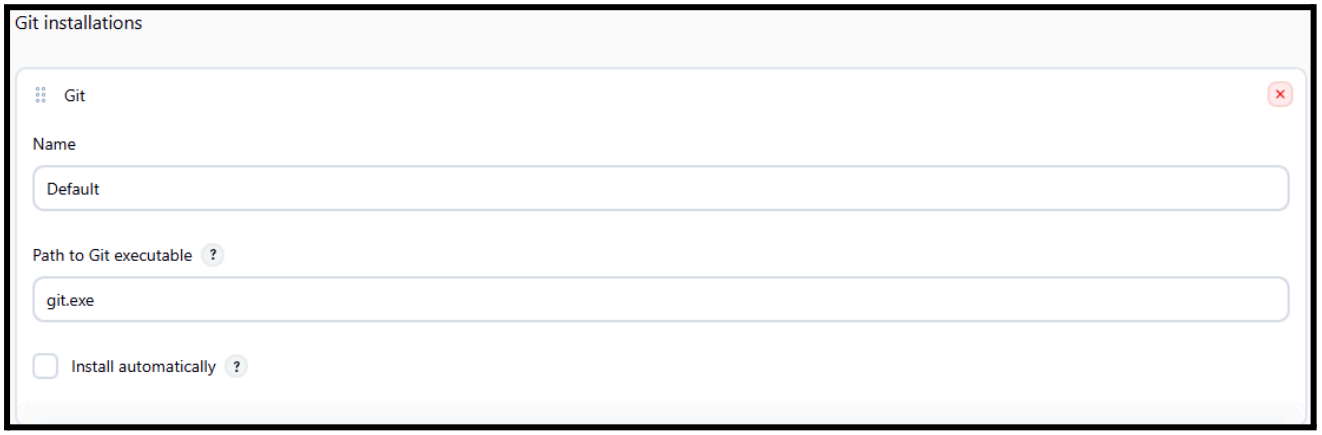
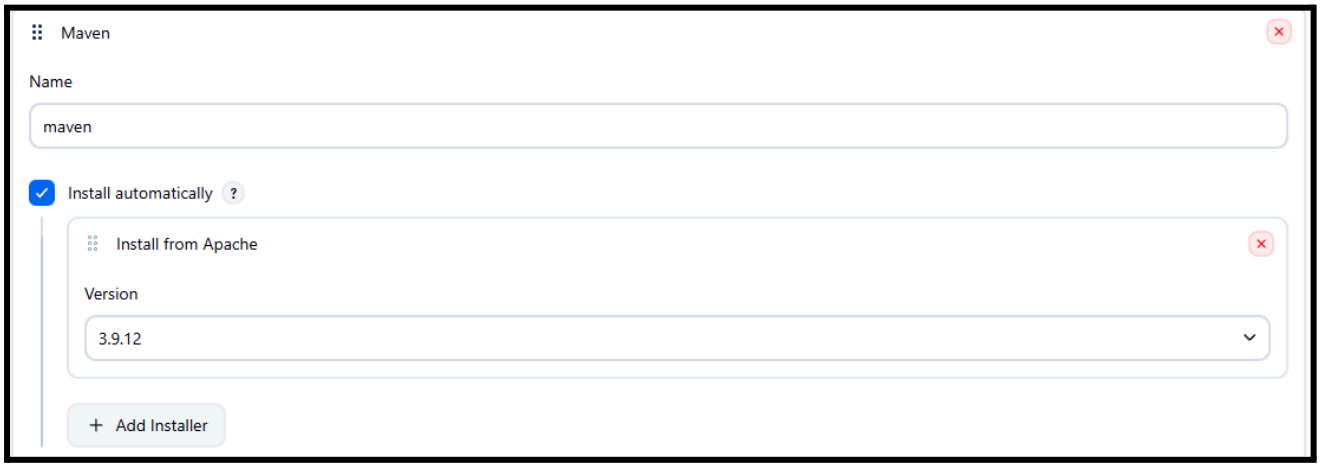
Configure Jenkins







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Jenkins / Manage Jenkins / Plugins

Plugins

Updates
Available plugins
Installed plugins
Advanced settings
Download progress

Build Pipeline

Install

Install	Name ↓	Released	Health
☑	HTML Publisher 427 Build Reports This plugin publishes HTML reports.	7 mo 13 days ago	93
☑	Maven Integration 3.27 Build Tools This plugin provides a deep integration between Jenkins and Maven. It adds support for automatic triggers between projects depending on SNAPSHOTs as well as the automated configuration of various Jenkins publishers such as Junit.	6 mo 3 days ago	100
☑	Python 1.3 Build Tools Adds the ability to execute python scripts as build steps. This plugin is deprecated. In general, this means that it is either obsolete, no longer being developed, or may no longer work. Learn more.	10 yr ago	31

Activate Windows
Go to Settings to activate Windows.

localhost:8081

Jenkins / Manage Jenkins / Plugins

Plugins

Updates
Available plugins
Installed plugins
Advanced settings
Download progress

Build Pipeline

Install

☑	Selenium 3.141.59 Cluster Management Warning: This plugin version may not be safe to use. Please review the following security notices: • Complete lack of CSRF protection can lead to OS command injection This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.	6 yr 9 mo ago	34
☑	Docker 1308.vff6e33248305 Cloud Providers · Cluster Management · docker This plugin integrates Jenkins with Docker	2 mo 22 days ago	100
☑	Build Pipeline 2.1.0 User Interface · Build Tools · Other Post-Build Actions		

Activate Windows
Go to Settings to activate Windows.

Jenkins / Manage Jenkins / Plugins

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Download progress

Build Pipeline

Install

	Adopt a Plugin initiative for more information.		
☑	Docker 1308.vff6e33248305 Cloud Providers · Cluster Management · docker This plugin integrates Jenkins with Docker	2 mo 22 days ago	100
☑	Build Pipeline 2.1.0 User Interface · Build Tools · Other Post-Build Actions This plugin renders upstream and downstream connected jobs that typically form a build pipeline. In addition, it offers the ability to define manual triggers for jobs that require intervention prior to execution, e.g. an approval process outside of Jenkins. This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.	1 mo 9 days ago	80

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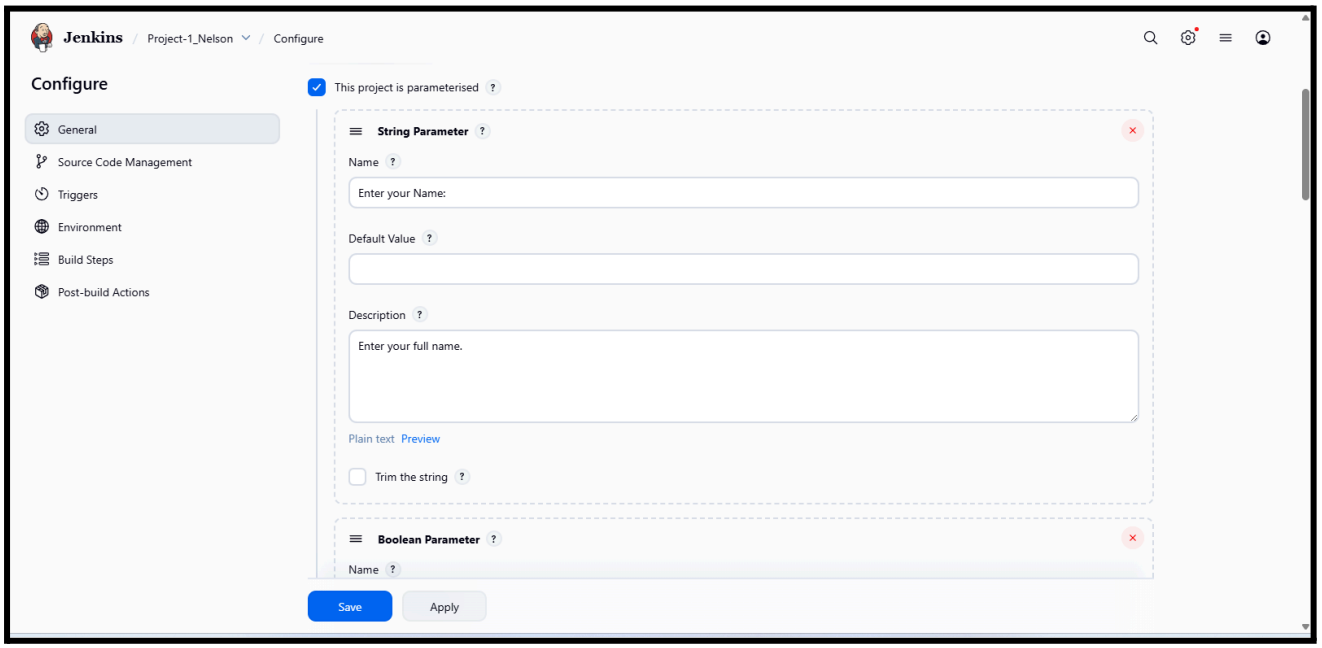
1. Freestyle project (Simple Windows batch command) – echo

2. Freestyle parameterized project (Windows batch command) – string, Boolean and choice parameters with echo



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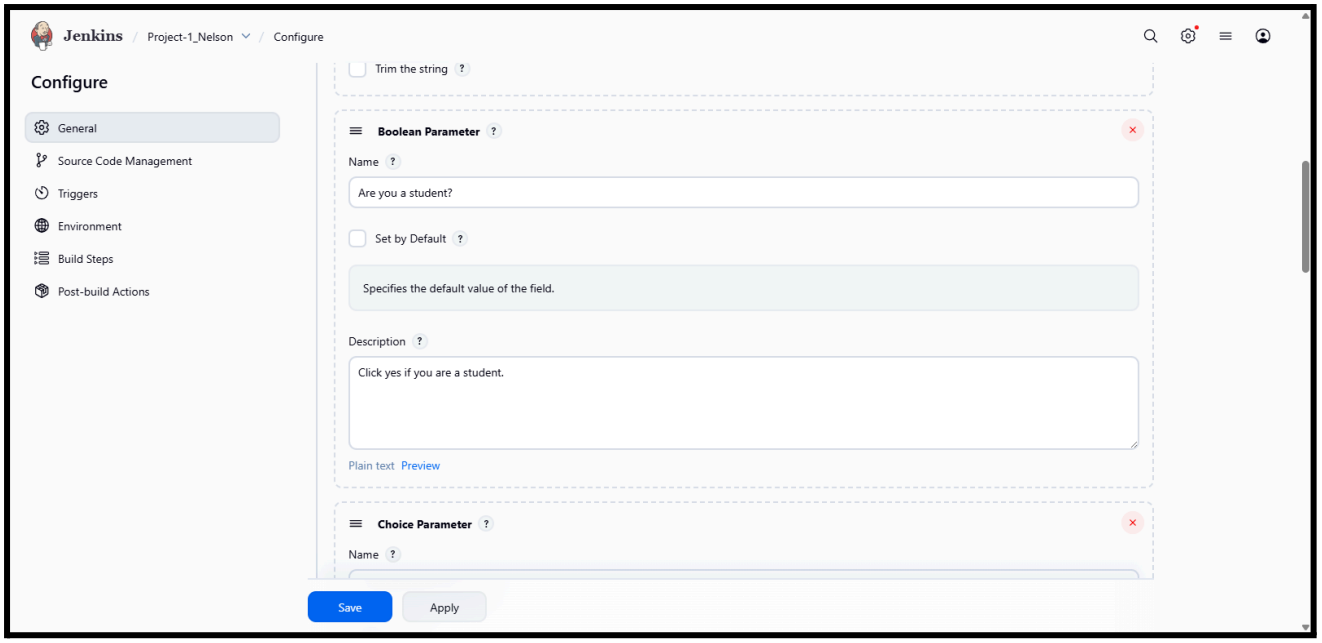

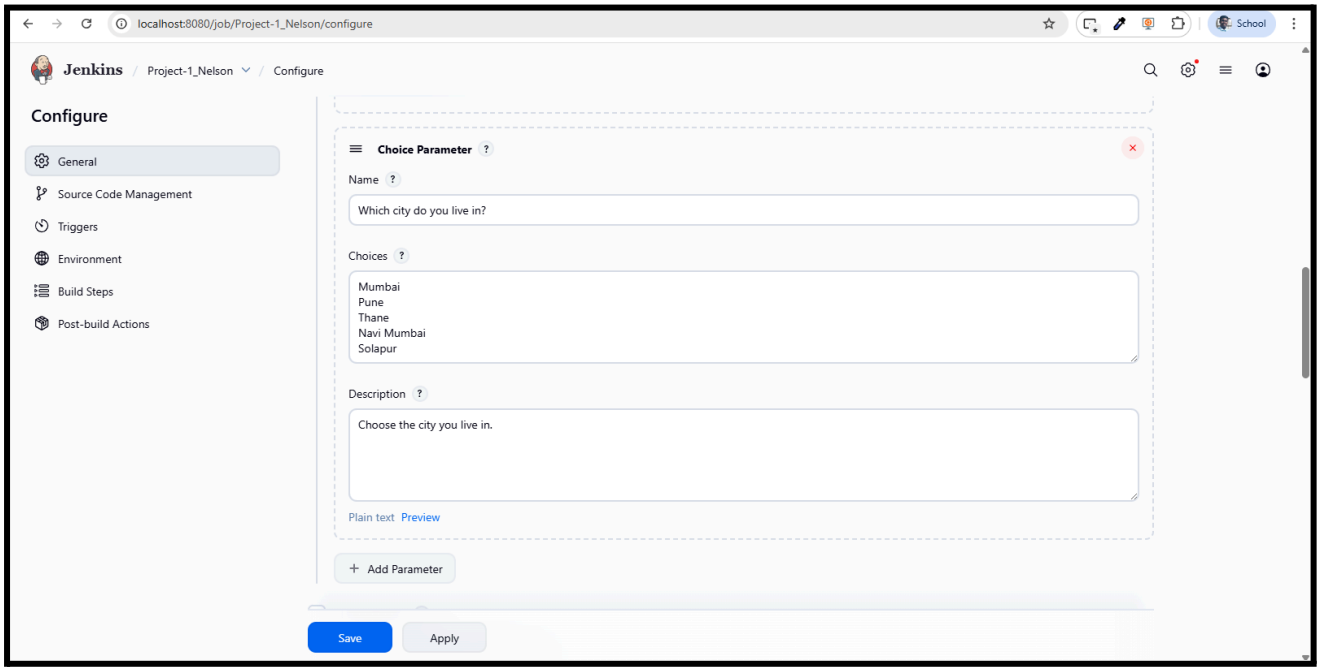

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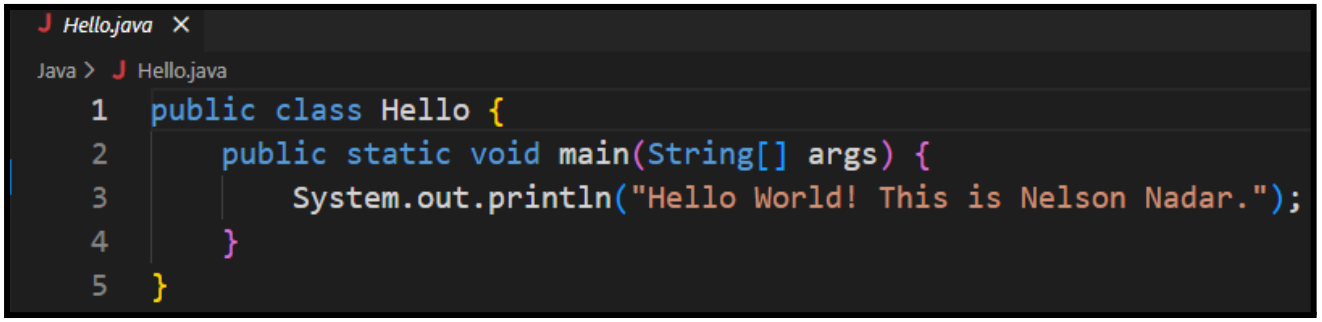
3. Java freestyle project - Hello world.



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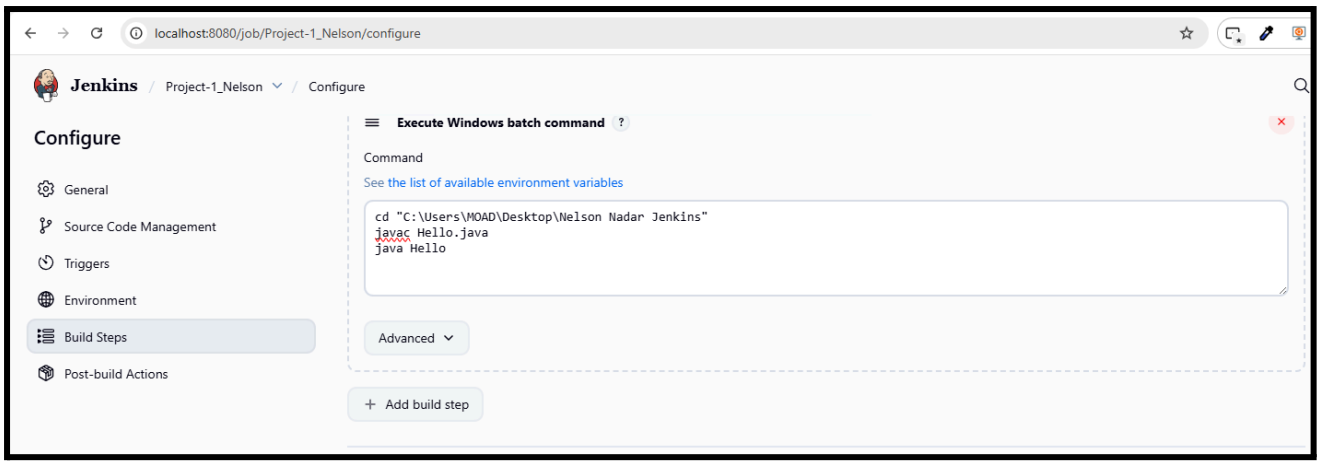

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4. Java freestyle project (parameterized) – 2 String parameters.

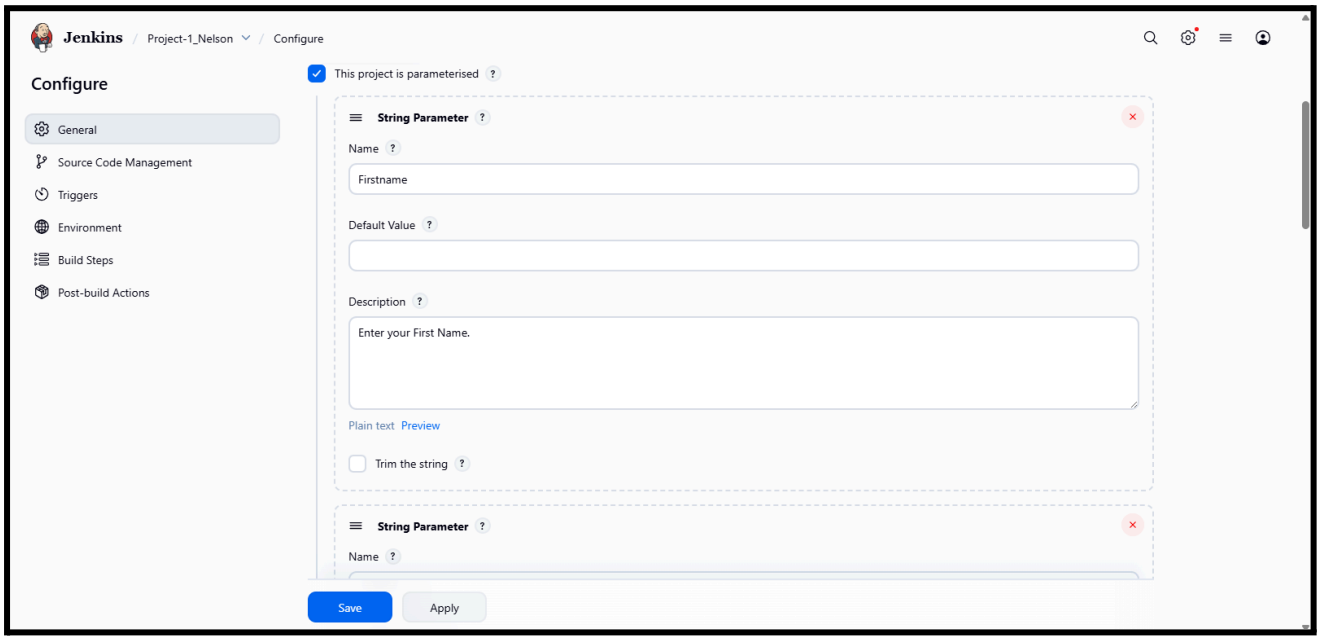
Name.java

Java > Name.java

```java
public class Name {
    public static void main(String[] args) {

        String firstName = System.getenv("Firstname");
        String lastName = System.getenv("Lastname");

        if (firstName == null || lastName == null) {
            System.out.println("Firstname or Lastname parameter is missing");
        } else {
            System.out.println("Firstname: " + firstName);
            System.out.println("Lastname: " + lastName);
        }
    }
}
```



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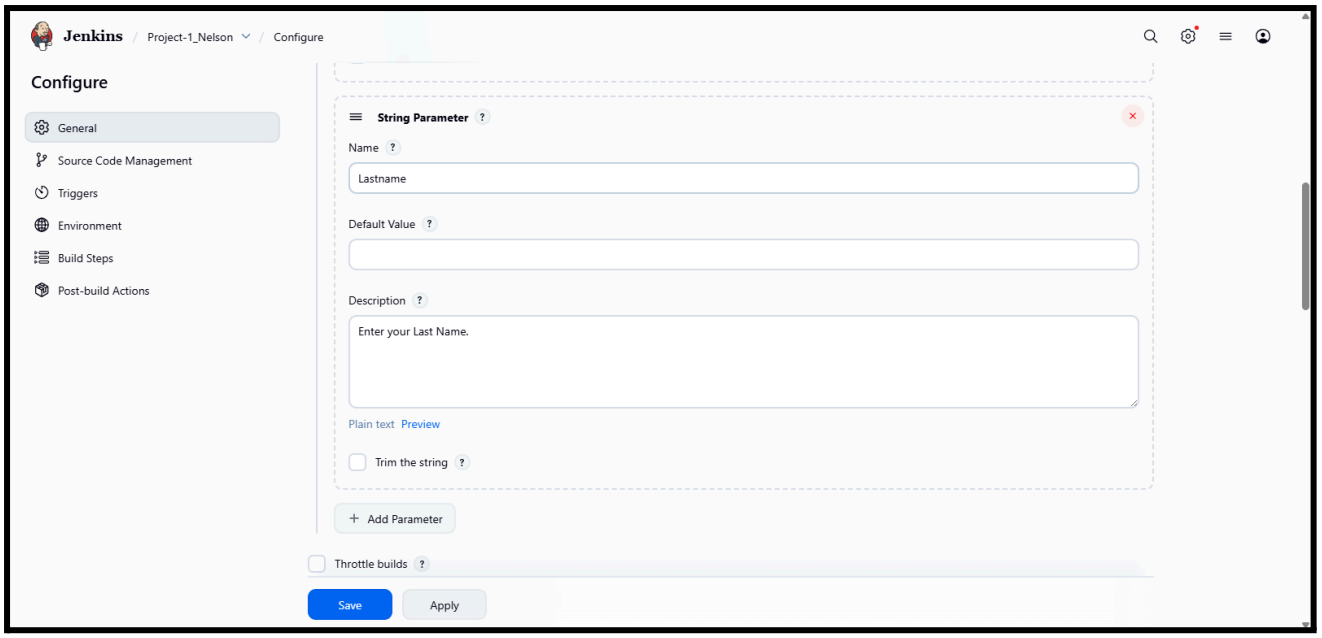

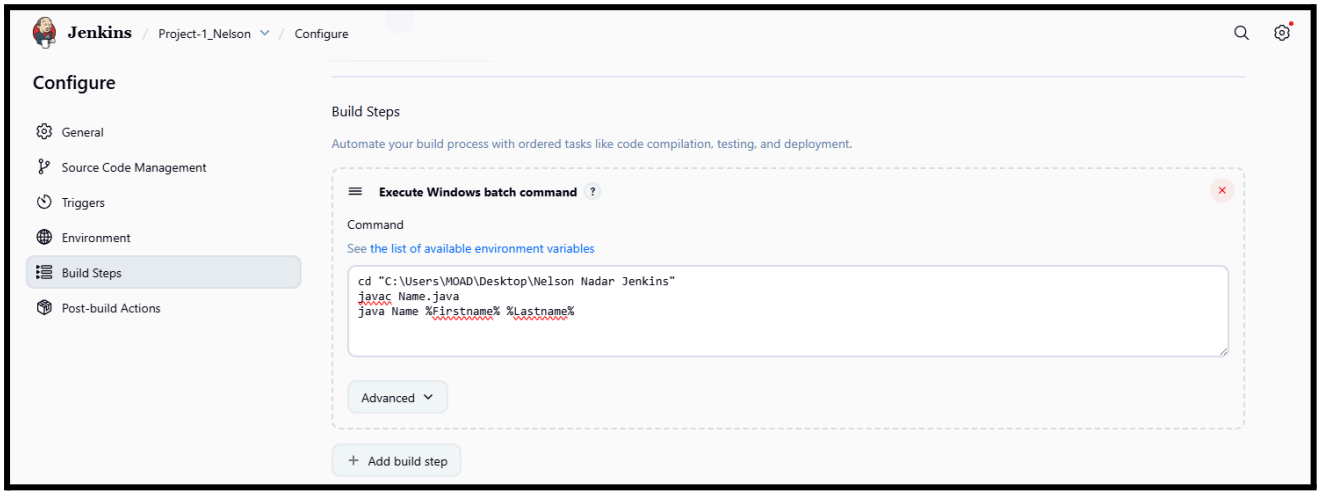

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